

Abdeltawwab Ahmed

AI Engineer

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SUMMARY

AI Engineer with strong foundations in Machine Learning and Deep Learning, with hands-on experience in Computer Vision, NLP, and Agentic AI systems. Skilled in building LLM-powered applications and intelligent workflows using frameworks like LangChain, LangGraph, and CrewAI, with practical experience in Hugging Face ecosystem and database integration. Focused on turning advanced AI concepts into scalable, real-world solutions.

PROFESSIONAL EXPERIENCE

ABS.AI, AI Engineer Intern

Apr 2026 – May 2026

Cairo, Egypt

- Contributed to the development of an AI-powered emotional storytelling SaaS platform using LLMs and expressive voice synthesis, enhancing interactive storytelling workflows.
- Built NLP pipelines and FastAPI-based services for real-time story generation, improving processing efficiency and reducing response latency by ~35%.

GlitchFix, Software Engineer

Feb 2025 – May 2025

Cairo, Egypt

- Developed responsive React-based admin dashboards, improving workflow usability and reducing UI inconsistencies across multiple modules.
- Built reusable UI components that accelerated frontend development and improved maintainability by reducing duplicated code by ~40%.

EDUCATION

National Telecommunications Institute, AI Scholarship (HireReady)

Oct 2025 – Jan 2026

- Built AI models for Vision and NLP using ML/DL techniques for real-world problems.
- Executed AI workflows including preprocessing, evaluation, and optimization across structured, image, and text data.

Faculty of computers and Artificial intelligence, Helwan University,

2020 – 2024

Bachelor's degree

Grade : Very Good

SKILLS

- **Programming Principles:** Problem solving | OOP | Database (Sql) | Data structures | Algorithms | Python
- **AI & ML Concepts:** EDA | Data Preprocessing | Machine Learning | Deep learning | NLP | Scikit-learn | Tensorflow | Pytorch | ETL (SSIS)
- **Agentic AI & LLMs:** AI Agents | GenAI | HuggingFace | LangChain | LangGraph | CrewAI | RAG | Fine-tuning | VLMs (OCR) | Vector databases
- **Deployment:** APIs (FastAPIs) | Git & GitHub | ngrok | Docker | VPS

PROJECTS

Smart Research Assistant Agent | [GitHub](#)

- Built an autonomous AI agent using LangChain v1 and LangGraph that orchestrates multi-tool workflows including web search, math evaluation, and note management with persistent memory across sessions.
- Implemented production-grade GenAI patterns including LCEL chain composition, RunnableParallel for concurrent LLM calls, Pydantic structured output, and ReAct tool-calling loop with a Gemini 2.5 Flash backbone.

Tech Stack: Python, LangChain v1, LangGraph, LCEL, Gemini 2.5 Flash, Pydantic, LangGraph Memory.

AI Storytelling SaaS Product — ABS.AI Company

- Contributed to the development of an AI-powered emotional storytelling SaaS platform using LLMs and expressive voice synthesis.
- Built NLP pipelines and RESTful APIs using FastAPI, Gemini, and AI workflow integrations for real-time story generation and narration.
- Collaborated on system integration, testing, and deployment preparation for scalable hosting environments.

Tech Stack: Python, NLP, LLMs, FastAPI, Gemini, Generative AI, Whisper.

Multi-Session AI Chatbot

- Built a conversational AI chatbot using LangChain and Gemini 2.5 Flash.
- Implemented session-based memory management with RunnableWithMessageHistory, enabling independent multi-session conversations and context retention.
- Designed dynamic prompt workflows using ChatPromptTemplate and message history injection.

Tech Stack: Python, LangChain, Gemini API, Prompt Engineering.

Speech Recognition System using Whisper, *ASR-Project-Using-Whisper*

- Developed a multilingual STT system using OpenAI's Whisper model for robust speech-to-text transcription across diverse audio conditions.
- Built and optimized audio processing pipeline for accurate transcription in real-world scenarios.

Tech Stack: Python, Deep Learning, NLP, Audio Processing.

Text-to-Speech System using VITS, *TTS-System-Vits-MetaAI*

- Built a text-to-speech (TTS) system using VITS from Meta AI to generate natural and high-quality speech from textual input.
- Leveraged end-to-end neural architectures for efficient and realistic voice synthesis.

Tech Stack: TTS, NLP, Audio Processing, Gradio.

Object Detection Assistant using DETR, *Object-Detection-Assistant*

- Developed an object detection system using DETR from Meta AI to detect objects and output their categories and counts.
- Built an end-to-end pipeline for image inference and structured result generation.

Tech Stack: Python, Deep Learning, Computer Vision, NLP, ViT.

LANGUAGES

Arabic — Native/Bilingual

English — Proficient